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Holcim Belgique SA
CEM V/A (S-V) 42,5 N-LH
HSR bulk "ECOPlanet"



ISSUED 08.12.2023
VALID UNTIL 08.12.2028

THIRD PARTY VERIFIED
in accordance with EN 15804+A2,
B-EPD Construction PCR
and EN 16908+A1

MODULES DECLARED

1 tonne (1000 kg) of CEM V/A (S-V) 42,5 N-LH HSR bulk "ECOPlanet".

Cradle-to-gate with options.

A123	A4	A5	B2 B4	C	D
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Dieter De Lathauwer

(Signature)

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The intended use of this EPD is to communicate scientifically based environmental information for construction products, for the purpose of assessing the environmental performance of buildings. This EPD is only valid when registered on www.b-epd.be. The FPS Public Health cannot be held responsible for the information provided by the owner of the EPD.



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1 PRODUCT DESCRIPTION

1.1 Product name

Commercial name: CEM V/A (S-V) 42,5 N-LH HSR bulk "ECOPlanet"

The exact (technical) identification of the product depends on the product certification:

- CE-mark: CEM V/A (S-V) 42,5 N-LH
- BENOR quality label: CEM V/A (S-V) 42,5 N-LH CE
HSR LA BENOR
- NF quality label: CEM V/A (S-V) 42,5 N-LH CE
PM-ES-CP1 NF
- KOMO quality label: /



1.2 Product description and intended use

Cement is a hydraulic binder, i.e. a finely ground inorganic material which, when mixed with water, forms a paste which sets and hardens by means of hydration reactions and processes and which, after hardening, retains its strength and stability even under water.

Cement is an intermediate product in the building sector and is mainly used as a binder for concrete, mortar or cement screed. CEM V/A (S-V) 42,5 N-LH HSR bulk "ECOPlanet" is a composite cement whose main constituents are Portland clinker, granulated blast-furnace slag (S) and siliceous fly ash (V). CEM V/A (S-V) 42,5 N-LH HSR is a low-heat cement according to EN 197-1, a sulfate-resisting cement according to NBN B12-108 and a cement with a limited alkali content according to NBN B12-109.

This EPD is a specific EPD of the CEM V/A (S-V) 42,5 N-LH HSR bulk "ECOPlanet" cement type that is produced both in Holcim Belgium's integrated cement plant in Obourg and in Holcim Belgium's cement blending plant in Antwerp.



1.3 Reference flow / declared unit

The product is delivered in bulk (trucks of 30 T or barges up to 1350 T).

The weight of the declared unit is 1000 kg (1 tonne or 1 T).

The absolute density of the product is 2910 kg / m³.

1.4 Installation

The installation of the product is not covered by the scope of the EPD as cement is an intermediate product.



1.5 Composition and content

Composition	Quantity
– Portland clinker produced in Obourg	44% to 54%
– Blast-furnace slag	18% to 28%
– Siliceous fly ash	16% to 26%
– Minor additional constituents	<= 5%
– Additives	<= 1%
– Gypsum and/or anhydrite	<= 6%

This EPD combines the impact from CEM V/A (S-V) 42,5 N-LH HSR bulk “ECOPlanet” produced both in Obourg and in Antwerp. Although the production processes are not the same (co-grinding of clinker and granulated slag with addition of fly ash to the separator in Obourg ; mixing of CEM I from Obourg with ground slag and fly ash sourced externally in Antwerp), the composition of the cement in both production locations is almost exactly identical (less than 2% for the main constituents). The impact from both production plants has been modelled separately but the results presented in this EPD are the combined results, weighed by the volumes expedited from both sites.

As for other EPD from Holcim Belgique, the data for the clinker – the main contributor to GWP – are average values from the reference year 2021. Since the specific cement type CEM V/A (S-V) 42,5 N-LH HSR bulk “ECOPlanet” has only been available for sale since early 2022, the cement composition has been taken as the average from 2022.

The product does not contain materials listed in the “Candidate list of Substances of Very High Concern for authorization”.

1.6 Reference service life

No RSL is declared for cement or building lime as they are intermediate building products (EN 16908+A1, 2022).

1.7 Description of geographical representativity

CEM V/A (S-V) 42,5 N-LH HSR bulk “ECOPlanet” is mainly sold to customers in Belgium, but exportations to France, Luxembourg, Germany and the Netherlands are common. Modules A1 to A3 are representative of the cement sold to all these countries. Module A4 is specific to the Belgian market. The EPD is representative for the Belgian market.

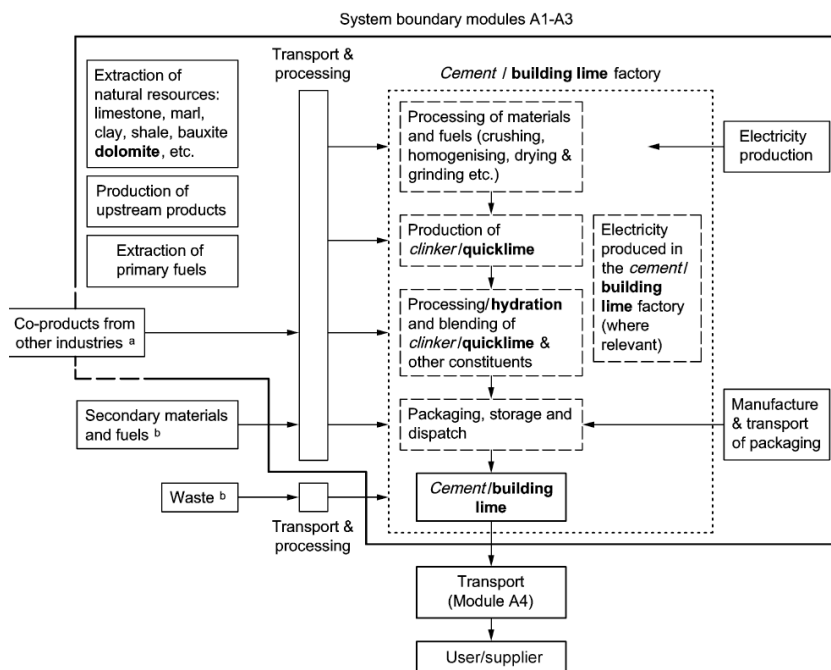


1.8 Description of the production process and technology

The cement-making process can be divided into two basic steps:

- Clinker (the main constituent of cement) is made in a kiln where the flame temperature reaches 2000°C, which heats raw materials such as limestone (calcium carbonate) with other silica, alumina and iron bearing materials (e.g. clay) up to 1450°C. During the first stage of this process, known as calcination, the calcium carbonate (limestone) is transformed into calcium oxide (free lime). During the second stage, known as clinkerisation, the calcium oxide reacts with the other constituents from the raw material mix to form new minerals, collectively called (Portland) clinker. This near-molten material is rapidly cooled to a temperature of 100 - 200°C.
- In a second step, clinker is ground together with gypsum and other materials such as limestone, blast furnace slag, siliceous fly ash, natural or calcined pozzolana (known as "MIC" Mineral Components or "SCM" Supplementary Cementitious Materials) to produce the fine (grey) powder known as cement. Other constituents are calcium sulfate (gypsum and/or anhydrite) to control setting, additives such as grindings aids and chromate reducing agents (the total quantity of which shall not exceed 1.0 % by mass of the cement) and minor additional constituents added to improve the physical properties of the cement, such as workability or water retention.

The diagram below describes the production of cement (taken from EN 16908):



Specific for the production facility in Antwerp, there is a third production step. The final cement is produced by blending (dry mixing) of different mineral powders: Portland cement CEM I 52,5 N from Obourg (as a source of the clinker, according to the 2 production steps described above), ground granulated blast furnace slag (as opposed to (coarse) granulated slag in Obourg that is co-ground with the clinker), siliceous fly ash, a small quantity of additional set regulator (anhydrite, as a complement to the set regulator already present in the CEM I) and minor additional constituents.



2 TECHNICAL DATA / PHYSICAL CHARACTERISTICS

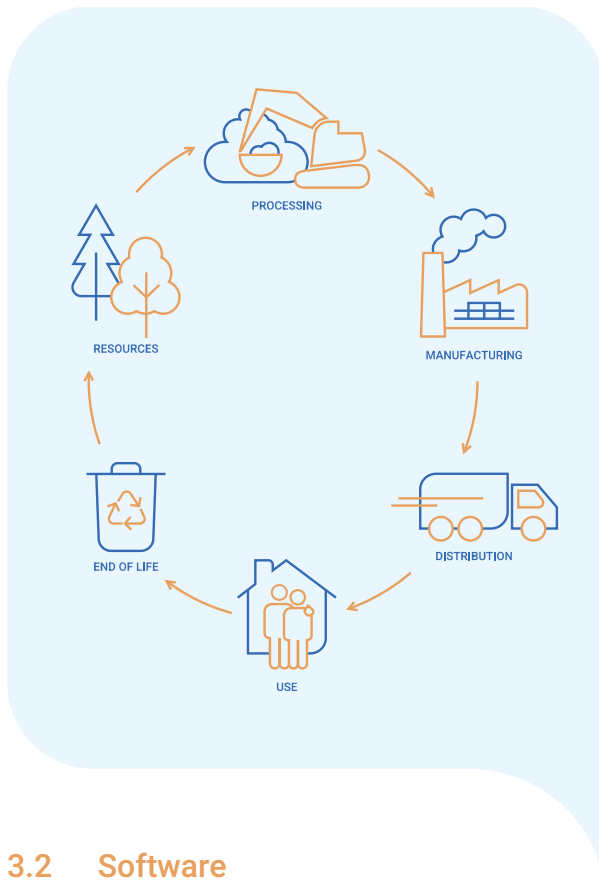
Technical property	Test standard	Specifications (EN 197-1)	Value	Unit
Standard consistence	EN 196-3	-	30	%
Initial setting time	EN 196-3	$\geq 01:00$	04:50	hh:mm
Final setting time	EN 196-3	$\leq 12:00$	05:50	hh:mm
Soundness	EN 196-3	≤ 10	< 1	mm
Compressive strength:				
2 days	EN 196-1	≥ 10	19	N/mm ²
28 days	EN 196-1	$\geq 42,5 / \leq 62,5$	53	N/mm ²
Fineness (Blaine)	EN 196-6	-	4300	cm ² /g
Density	EN 196-6	-	2910	kg/m ³



3 LCA-STUDY

3.1 Date of LCA-study

September 27, 2023



3.2 Software

For the calculation of the LCA results, the software program "GCCA EPD Tool" (v4.0) was used for A1 to A3 calculations and is based on ecoinvent 3.5. The "GCCA EPD Tool" is a specific tool to the cement and concrete industry, it is compliant with the EN 15804+A2 standard and is verified by EPD international. SimaPro was used to model module A4 module in combination with ecoinvent v3.8.

3.3 Information on allocation

The allocation rules defined by EN 15804+A2 and EN16908 have been applied.

The manufacturing process does not produce co-products that require allocation.

A cut-off is applied when recyclable waste reaches the end of the waste state.

By default, this EPD addresses the environmental impacts of cement production excluding the incineration of the input waste (net figure). For clarity and transparency, the gross figure is also communicated as recommended in the EN 16908+A1.

For the blast furnace slag, a co-product of the pig iron production, economic allocation has been used conform EN 15804+A2, based on EPD data from the main supplier.

For the fly ash, a waste from coal-fired power plants, no allocated has been accounted for in this EPD.

3.4 Information on cut off

The following processes are considered below cut-off:

- Cleaning of the installation
- Wear of grinding balls
- Cement grinding losses
- Cement mixing losses

3.5 Information on excluded processes

Following processes were excluded for the inventory:

- Production and construction of equipment for buildings and installations
- Production and construction of equipment for the production of clinker and cement and lab analyses
- Activities related to the personnel (mobility, furniture and desk consumables)
- Utilization of water and energy related to the sales and management of the product
- Cleaning of offices



3.6 Information on biogenic carbon modelling

The product does not contain biogenic carbon. The product is sold in bulk and does not require packaging.

Biogenic carbon content	(kg C / FU)
Biogenic carbon content in product (at the gate)	0
Biogenic carbon content in accompanying packaging (at the gate)	0

3.7 Information on carbon offsetting

Carbon offsetting is not allowed in the EN 15804+A2 and hence not taken into account in the calculations

3.8 Information on carbonation of cementitious materials

No carbonation of cementitious materials is considered in the scope of this EPD. The construction products made with cement (mortar, concrete) may indeed carbonate in the following life cycle stages.

3.9 Additional or deviating characterisation factors

The EF3.0 version of the EN 15804+A2 LCIA method was used. The characterization factors from EC-JRC were applied. No additional or deviating characterisation factors were used.

3.10 Description of the variability

The EPD is specific to the product and manufacturing site. No underlying parameters or assumptions that would impact the results were made. The results do not vary.



3.11 Specificity

The data used for the LCA are specific for this product which is manufactured by a single manufacturer in a single production site.

3.12 Period of data collection

Manufacturer specific data have been collected for the year 2021.

3.13 Information on data collection

The activity data is based on site specific data (volumes, lower heating values, emission factors) verified by the EU ETS system for the clinker production. For the cement production, the main data used is the recipe.

Background data is included in the GCCA EPD tool and has been verified by EPD International.

3.14 Database used for background data

Ecoinvent v3.5 for A1 to A3 and ecoinvent version 3.8 for A4.

3.15 Energy mix

The residual electricity mix of Belgium in 2021 was used, based on the electricity attributes provided by the Association of Issuing Bodies (AIB).



4 PRODUCTION SITES

The product is produced both in Obourg, where Holcim Belgium operates an integrated cement plant (photo on the left), as in Antwerp, where Holcim Belgium operates a cement blending plant (photo on the right).



5 SYSTEM BOUNDARIES

Product stage			Construction installation stage		Use stage							End of life stage				Beyond the system boundaries
Raw materials	Transport	Manufacturing	Transport	Construction installation stage	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
☒	☒	☒	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

X = included in the EPD

□ = module not declared



6 POTENTIAL ENVIRONMENTAL IMPACTS PER REFERENCE FLOW

Reference results: **Net** impacts for 1 ton of cement production.

	Production	Construction process stage		Use stage							End-of-life stage				D Reuse, recovery, recycling
	A1-A3 Product stage	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
 GWP total (kg CO2 equiv/FU)	4,02E+02	1,62E+01													
 GWP fossil (kg CO2 equiv/FU)	4,02E+02	1,62E+01													
 GWP biogenic (kg CO2 equiv/FU)	4,33E-02	1,49E-02													
 GWP luluc (kg CO2 equiv/FU)	3,66E-02	6,31E-03													
 ODP (kg CFC 11 equiv/FU)	1,68E-05	3,77E-06													
 AP (mol H+ eq/FU)	2,01E+00	6,61E-02													
 EP - freshwater (kg (PO4)3- equiv/FU)	1,89E-02	1,04E-03													
 EP - marine (kg (PO4)3- equiv/FU)	7,61E-03	1,99E-02													
 EP - terrestrial (kg (PO4)3- equiv/FU)	4,82E+00	2,18E-01													
 POCP (kg Ethene equiv/FU)	1,17E+00	6,72E-02													
 ADP Elements (kg Sb equiv/FU)	1,78E-04	5,40E-05													
 ADP fossil fuels (MJ/FU)	2,72E+03	2,46E+02													
 WDP (m³ water eq deprived /FU)	2,00E+01	7,55E-01													

Additional insight for transparency: **Gross** impacts for 1 ton of cement

Production		Construction process stage		Use stage							End-of-life stage				
A1-A3 Product stage		A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
	GWP total (kg CO2 equiv/FU)	5,18E+02	1,62E+01												
	GWP fossil (kg CO2 equiv/FU)	5,18E+02	1,62E+01												
	GWP biogenic (kg CO2 equiv/FU)	1,30E-01	1,49E-02												
	GWP luluc (kg CO2 equiv/FU)	3,53E-02	6,31E-03												
	ODP (kg CFC 11 equiv/FU)	1,59E-05	3,77E-06												
	AP (mol H+ eq/FU)	2,00E+00	6,61E-02												
	EP - freshwater (kg (PO4)3- equiv/FU)	1,83E-02	1,04E-03												
	EP - marine (kg (PO4)3- equiv/FU)	7,50E-03	1,99E-02												
	EP - terrestrial (kg (PO4)3- equiv/FU)	4,80E+00	2,18E-01												
	POCP (kg Ethene equiv/FU)	1,17E+00	6,72E-02												
	ADP Elements (kg Sb equiv/FU)	1,70E-04	5,40E-05												
	ADP fossil fuels (MJ/FU)	2,61E+03	2,46E+02												
	WDP (m³ water eq deprived /FU)	1,89E+01	7,55E-01												

GWP TOTAL = TOTAL GLOBAL WARMING POTENTIAL (CLIMATE CHANGE); GWP-LULUC = GLOBAL WARMING POTENTIAL (CLIMATE CHANGE) LAND USE AND LAND USE CHANGE; ODP = OZONE DEPLETION POTENTIAL; AP = ACIDIFICATION POTENTIAL FOR SOIL AND WATER; EP = EUTROPHICATION POTENTIAL; POCP = PHOTOCHEMICAL OZONE CREATION; ADPE = ABIOTIC DEPLETION POTENTIAL – ELEMENTS; ADPF = ABIOTIC DEPLETION POTENTIAL – FOSSIL FUELS; WDP = WATER USE (WATER (USER) DEPRIVATION POTENTIAL, DEPRIVATION-WEIGHTED WATER CONSUMPTION)

7 RESOURCE USE

	Production	Construction process stage		Use stage							End-of-life stage				D Reuse, recovery, recycling
	A1-A3 Product stage	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
PERE (MJ/FU, net calorific value)	1,64E+02	7,73E-01													
PERM (MJ/FU, net calorific value)	0,00E+00	0,00E+00													
PERT (MJ/FU, net calorific value)	1,64E+02	7,73E-01													
PENRE (MJ/FU, net calorific value)	2,76E+03	2,24E+02													
PENRM (MJ/FU, net calorific value)	0,00E+00	0,00E+00													
PENRT (MJ/FU, net calorific value)	2,76E+03	2,24E+02													
SM (kg/FU)	5,29E+02	0,00E+00													
RSF (MJ/FU, net calorific value)	4,72E+02	0,00E+00													
NRSF (MJ/FU, net calorific value)	1,29E+03	0,00E+00													
FW (m³ water eq/FU)	6,26E+00	2,52E-02													

PERE = USE OF RENEWABLE PRIMARY ENERGY EXCLUDING RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PERM = USE OF RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PERT = TOTAL USE OF RENEWABLE PRIMARY ENERGY RESOURCES; PENRE = USE OF NON-RENEWABLE PRIMARY ENERGY EXCLUDING NON-RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PENRM = USE OF NON-RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PENRT = TOTAL USE OF NON-RENEWABLE PRIMARY ENERGY RESOURCES; SM = USE OF SECONDARY MATERIAL; RSF = USE OF RENEWABLE SECONDARY FUELS; NRSF = USE OF NON-RENEWABLE SECONDARY FUELS; FW = NET USE OF FRESH WATER

8 WASTE CATEGORIES & OUTPUT FLOWS

	Production	Construction process stage		Use stage							End-of-life stage				D Reuse, recovery, recycling
	A1-A3 Product stage	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
Hazardous waste disposed (kg/FU)	1,08E-04	0,00E+00													
Non-hazardous waste disposed (kg/FU)	2,42E+00	0,00E+00													
Radioactive waste disposed (kg/FU)	8,50E-03	0,00E+00													
Components for re-use (kg/FU)	0,00E+00	0,00E+00													
Materials for recycling (kg/FU)	0,00E+00	0,00E+00													
Materials for energy recovery (kg/FU)	0,00E+00	0,00E+00													
Exported energy (MJ/FU)	0,00E+00	0,00E+00													

9 IMPACT CATEGORIES ADDITIONAL TO EN 15804

		Production	Construction process stage		Use stage							End-of-life stage				
		A1-A3 Product stage	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
	PM (disease incidence)	1,79E-05	1,47E-06													
	IRHH (kg U235 eq/FU)	5,40E+04	1,26E+00													
	ETF (CTUe/FU)	1,07E+02	1,92E+02													
	HTCE (CTUh/FU)	1,40E-06	6,11E-09													
	HTnCE (CTUh/FU)	4,24E-05	2,03E-07													
	Land Use Related impacts (dimensionless)	1,04E+03	1,85E+02													

HTCE = HUMAN TOXICITY – CANCER EFFECTS; HTNCE = HUMAN TOXICITY – NON CANCER EFFECTS; ETF = ECOTOXICITY – FRESHWATER; (POTENTIAL COMPARATIVE TOXIC UNIT)

PM = PARTICULATE MATTER (POTENTIAL INCIDENCE OF DISEASE DUE TO PM EMISSIONS);

IRHH = IONIZING RADIATION – HUMAN HEALTH EFFECTS (POTENTIAL HUMAN EXPOSURE EFFICIENCY RELATIVE TO U235);

9.1 Environmental impact categories explained

	Global Warming Potential	The global warming potential of a gas refers to the total contribution to global warming resulting from the emission of one unit of that gas relative to one unit of the reference gas, carbon dioxide, which is assigned a value of 1. It is split up in 4: – Global Warming Potential total (GWP-total) which is the sum of GWP-fossil, GWP-biogenic and GWP-luluc – Global Warming Potential fossil fuels (GWP-fossil) : The global warming potential related to greenhouse gas (GHG) emissions to any media originating from the oxidation and/or reduction of fossil fuels by means of their transformation or degradation (e.g. combustion, digestion, landfilling, etc). – Global Warming Potential biogenic (GWP-biogenic) : The global warming potential related to carbon emissions to air (CO₂, CO and CH₄) originating from the oxidation and/or reduction of aboveground biomass by means of its transformation or degradation (e.g. combustion, digestion, composting, landfilling) and CO₂ uptake from the atmosphere through photosynthesis during biomass growth - i.e. corresponding to the carbon content of products, biofuels or above ground plant residues such as litter and dead wood. – Global Warming Potential land use and land use change (GWP-luluc): The global warming potential related to carbon uptakes and emissions (CO₂, CO and CH₄) originating from carbon stock changes caused by land use change and land use. This sub-category includes biogenic carbon exchanges from deforestation, road construction or other soil activities (including soil carbon emissions).
	Ozone Depletion	Destruction of the stratospheric ozone layer which shields the earth from ultraviolet radiation harmful to life. This destruction of ozone is caused by the breakdown of certain chlorine and/or bromine containing compounds (chlorofluorocarbons or halons), which break down when they reach the stratosphere and then catalytically destroy ozone molecules.
	Acidification potential	Acid depositions have negative impacts on natural ecosystems and the man-made environment incl. buildings. The main sources for emissions of acidifying substances are agriculture and fossil fuel combustion used for electricity production, heating and transport.
	Eutrophication potential	The potential to cause over-fertilization of water and soil, which can result in increased growth of biomass and following adverse effects. It is split up in 3: – Eutrophication potential - freshwater: The potential to cause over-fertilization of freshwater, which can result in increased growth of biomass and following adverse effects. – Eutrophication potential - marine: The potential to cause over-fertilization of marine water, which can result in increased growth of biomass and following adverse effects. – Eutrophication potential - terrestrial: The potential to cause over-fertilization of soil, which can result in increased growth of biomass and following adverse effects.
	Photochemical ozone creation	Chemical reactions brought about by the light energy of the sun creating photochemical smog. The reaction of nitrogen oxides with hydrocarbons in the presence of sunlight to form ozone is an example of a photochemical reaction.
	Abiotic depletion potential for non-fossil resources	Consumption of non-renewable resources, thereby lowering their availability for future generations. Expressed in comparison to Antimony (Sb). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Abiotic depletion potential for fossil resources	Measure for the depletion of fossil fuels such as oil, natural gas, and coal. The stock of the fossil fuels is formed by the total amount of fossil fuels, expressed in Megajoules (MJ). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.



	Ecotoxicity for aquatic fresh water	The impacts of chemical substances on ecosystems (freshwater). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Human toxicity (carcinogenic effects)	The impacts of chemical substances on human health via three parts of the environment: air, soil and water. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Human toxicity (non-carcinogenic effects)	The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Particulate matter	Accounts for the adverse health effects on human health caused by emissions of Particulate Matter (PM) and its precursors (NO_x, SO_x, NH₃)
	Resource depletion (water)	Accounts for water use related to local scarcity of water as freshwater is a scarce resource in some regions, while in others it is not. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Ionizing radiation - human health effects	This impact category deals mainly with the eventual impact on human health of low dose ionizing radiation of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.
	Land use related impacts	The indicator is the "soil quality index" which is the result of an aggregation of following four aspects: – Biotic production – Erosion resistance – Mechanical filtration – Groundwater The aggregation is done based on a JRC model. The four aspects are quantified through the LANCA model for land use. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.



10 DETAILS OF THE UNDERLYING SCENARIOS USED TO CALCULATE THE IMPACTS

10.1 A1 - raw material supply

This module takes into account the extraction and processing of all raw materials and energy which occur upstream to the studied manufacturing process.

The blast furnace slag input has been economically allocated as a co-product of pig iron production.

The impact of the siliceous fly ash (a waste from coal-fire electricity production) has been allocated to the previous system.

10.2 A2 – transport to the manufacturer

The raw materials are transported to the manufacturing site. The supplier specific distances have been used.

10.3 manufacturing

This module takes into account the production process, no assumptions have been made.

10.4 A4 – transport to the building site

Fuel type and consumption of vehicle or vehicle type used for transport	Truck 16-32 ton 0,037 kg diesel / tkm	Truck >32 ton 0,019 kg diesel / tkm	Truck 7.5-16 ton 0,047 kg diesel / tkm
Distance	100	100	35
Capacity utilisation (including empty returns)	Default ecoinvent	Default ecoinvent	Default ecoinvent
Bulk density of transported products	-	-	-
Volume capacity utilisation factor	Default ecoinvent	Default ecoinvent	Default ecoinvent

The B-PCR provides default transport scenarios for the transport to the building site for cases where specific data on transport are missing. The B-PCR provides scenarios for this life cycle stage. The following transport steps apply:

- 75% directly to the construction site over 100 km with a Lorry 16-32 ton (EURO5) (ecoinvent record: 'Transport, freight, lorry 16-32 metric ton, EURO5 {RER}| transport, freight, lorry 16-32 metric ton, EURO5 | Cut-off, U')
- 25% to a supplier over 100 km with a >32 ton lorry (ecoinvent record: 'Transport, freight, lorry >32 metric ton, EURO5 {RER}| transport, freight, lorry >32 metric ton, EURO5 | Cut-off, U'). Afterwards the product is transported from the supplier to the construction site. 90% of the 25% is transported with a Lorry 16-32 ton (EURO5) over a distance of 35km and 10% of the 25% is transported with a Lorry 7.5-16 ton (EURO5) over a distance of 35km. (ecoinvent records: 'Transport, freight, lorry 16-32 metric ton, EURO5 {RER}| transport, freight, lorry 16-32 metric ton, EURO5 | Cut-off, U', 'Transport, freight, lorry 7.5-16 metric ton, EURO5 {RER}| transport, freight, lorry 7.5-16 metric ton, EURO5 | Cut-off, U')



10.5 A5 – installation in the building

Not applicable for this EPD.

10.6 B – use stage (excluding potential savings)

Not applicable for this EPD.

10.7 C: End of life

Not applicable for this EPD.

10.8 D – Benefits and loads beyond the system boundaries

Not applicable for this EPD.



11 RELEASE OF DANGEROUS SUBSTANCES TO INDOOR AIR, SOIL AND WATER DURING THE USE STAGE

11.1 Indoor air

Cement is not in direct contact with indoor air. It is the cement-based products (concrete, mortar ...) that are potentially in contact with indoor air.

11.2 Soil and water

Cement is not in direct contact with soil and water. It is the cement-based products (concrete, mortar ...) that are potentially in contact with soil and water.

12 DEMONSTRATION OF VERIFICATION

EN 15804+A2 serves as the core PCR

Independent verification of the environmental declaration and data according to standard EN ISO 14025:2010

Internal ☐ External ☒

Third party verifier: [Evert Vermaut, Jan Olieslagerslaan 35, 1800 Vilvoorde, rsterckx@vincotte.be]

13 LCA INTERPRETATION

Not provided.



14 TECHNICAL INFORMATION FOR SCENARIO DEVELOPMENT

Not applicable for this EPD.

15 APPLICATION UNIT

Not applicable for this EPD.

16 ADDITIONAL INFORMATION ON REVERSIBILITY

Not applicable for this EPD.



17 BIBLIOGRAPHY

ISO 14040:2006: Environmental Management-Life Cycle Assessment-Principles and framework.
ISO 14044:2006: Environmental Management-Life Cycle Assessment-Requirements and guidelines.
ISO 14025:2006: Environmental labels and Declarations-Type III Environmental Declarations-Principles and procedures.
NBN EN 15804+A2:2019: Sustainability of construction works – Environmental product declarations – Core rules for the product category of construction products.
NBN EN 16908+A1:2022: Cement and building lime - Environmental product declarations - Product category rules complementary to EN 15804
B-EPD Construction Product Category Rules complementary to NBN EN 15804+A2 v18.10.2022

General information

Owner of the EPD, Responsible for the data,
LCA and information



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Project report: CEM V/A (S-V) 42,5 N-LH HSR bulk "ECOPlanet"
EPD Background report



Verifier

Evert Vermaut Vinçotte nv
Date of verification: 30.10.2023
External independent verification of the declaration and data
according to EN ISO 14025 and relevant PCR documents

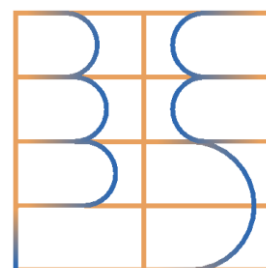
Comparing EPDs is not possible unless they are conform to the same PCR and taking into account the building context.
The program operator cannot be held responsible for the information supplied by the owner of the EPD nor LCA practitioner.



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